Evaluating Trigonometric Functions

Name _____

Algebra 2 Lesson 7-2 (fast track) Date _____ Period _____

Learning goal: I can use special triangles and reference angles to find the values of all six trigonometric functions of an angle in any quadrant.

Language goal: I can explain how I found a trigonometric value using the words reference angle, quadrant, terminal point, and reciprocal.

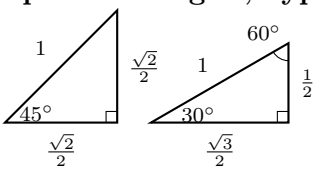
Spanish: *ángulo de referencia; secante, cosecante, cotangente; funciones trigonométricas recíprocas.*

On the quiz: (1) Sine and cosine of a special angle such as $\frac{2\pi}{3}$ (Example 4). (2) Sine from cosine (full Savvas quiz only). (3) Bonus: which statements about special angles are true, including negative angles and angles past 2π (Examples 4 and 5). (4) Cosecant, cotangent and secant of 225° (Example 5 and the Practice). (5) A point on a radius-8 circle (full Savvas quiz only).

Example 4: Evaluate Trigonometric Functions with Special Angles as Reference Angles

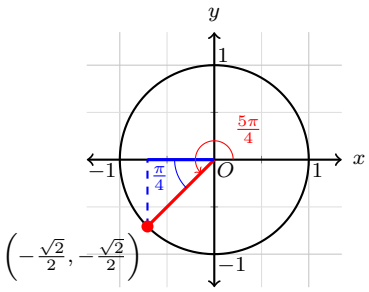
1. Find the **quadrant** of the terminal side.
2. Find the **reference angle**: the acute angle between the terminal side and the x-axis.
3. A special triangle (hypotenuse 1) gives the legs; the quadrant gives the signs. That is the **terminal point** $(x, y) = (\cos\theta, \sin\theta)$.
4. $\sin\theta = y$, $\cos\theta = x$, $\tan\theta = \frac{y}{x}$.

Special triangles, hypotenuse 1 (Example 3)



30°-60°-90°: short leg = half the hypotenuse; long leg = short leg $\times\sqrt{3}$.

A. What are the sine, cosine, and tangent of $\frac{5\pi}{4}$?

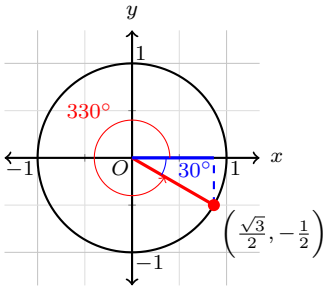


Quadrant III. Reference angle $\frac{5\pi}{4} - \pi = \frac{\pi}{4}$, or 45°.

$\sin\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}$, $\cos\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}$

$\tan\frac{5\pi}{4} = \frac{-\sqrt{2}/2}{-\sqrt{2}/2} = 1$

B. What are the sine, cosine, and tangent of 330°?

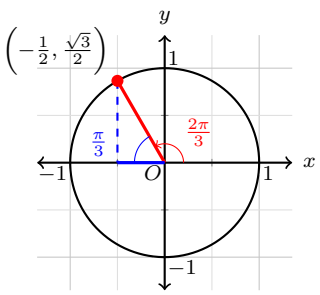


Quadrant IV. Reference angle $360° - 330° = 30°$, or $\frac{\pi}{6}$.

$\sin 330° = -\frac{1}{2}$, $\cos 330° = \frac{\sqrt{3}}{2}$

$\tan 330° = \frac{-1/2}{\sqrt{3}/2} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$

C. What are the sine, cosine, and tangent of $\frac{2\pi}{3}$?



Quadrant II. Reference angle $\pi - \frac{2\pi}{3} = \frac{\pi}{3}$, or 60°.

$\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$, $\cos\frac{2\pi}{3} = -\frac{1}{2}$

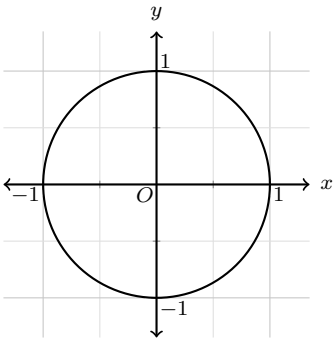
$\tan\frac{2\pi}{3} = \frac{\sqrt{3}/2}{-1/2} = -\sqrt{3}$

Say it: The terminal side of 330° is in Quadrant _____, so the reference angle is _____. The terminal point is _____, and $\sin 330°$ is negative because

Try It 4: Find the sine, cosine, and tangent of angles $\frac{4\pi}{3}$ and $-210°$

Hint: Sketch each angle and its reference triangle on the unit circle, then fill in the steps. A negative angle turns clockwise.

a. $\theta = \frac{4\pi}{3}$



Quadrant: _____

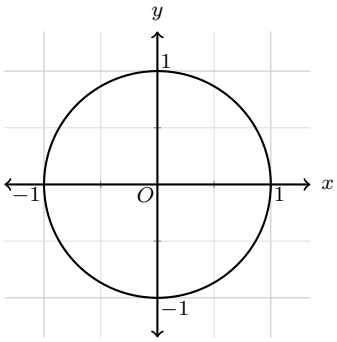
Reference angle: _____

Terminal point: (_____, _____)

$\sin\frac{4\pi}{3} =$ _____ $\cos\frac{4\pi}{3} =$ _____

$\tan\frac{4\pi}{3} =$ _____

b. $\theta = -210°$



Quadrant: _____

Reference angle: _____

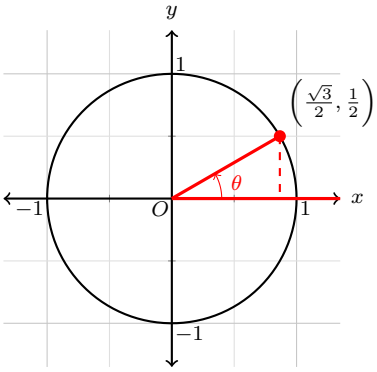
Terminal point: (_____, _____)

$\sin(-210°) =$ _____ $\cos(-210°) =$ _____

$\tan(-210°) =$ _____

Example 5: Use Reciprocal Trigonometric Identities

What are the six basic trigonometric functions for angle θ? In addition to sine, cosine, and tangent, there are three more ratios that are defined as **reciprocal** trigonometric functions.



Here $\theta = \frac{\pi}{6}$ (30°).

Cosecant: $\csc\theta = \dfrac{1}{\sin\theta} = \dfrac{1}{y}$ Secant: $\sec\theta = \dfrac{1}{\cos\theta} = \dfrac{1}{x}$

Cotangent: $\cot\theta = \dfrac{1}{\tan\theta}$, also $\cot\theta = \dfrac{\cos\theta}{\sin\theta} = \dfrac{x}{y}$

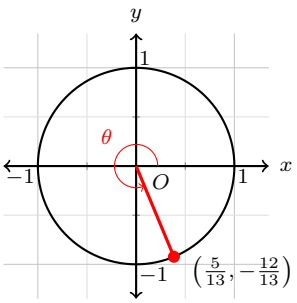
$\sin\theta = \frac{1}{2}$	$\csc\theta = \frac{2}{1} = 2$
$\cos\theta = \frac{\sqrt{3}}{2}$	$\sec\theta = \frac{2}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$
$\tan\theta = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{3}}{3}$	$\cot\theta = \frac{3}{\sqrt{3}} = \frac{3\sqrt{3}}{3} = \sqrt{3}$

Check: $\cot\theta = \dfrac{x}{y} = \dfrac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$. ✓ A reciprocal has the **same sign** as its function, so the quadrant signs from Example 4 carry over.

Say it: The secant is the _____ of the cosine, so for $\theta = \frac{\pi}{6}$, $\sec\theta =$ _____. To find $\csc\theta$ I use the _____-coordinate of the terminal point because

Try It 5: Write the six trigonometric functions for the given angle θ

a.



$\sin\theta =$ _____

$\cos\theta =$ _____

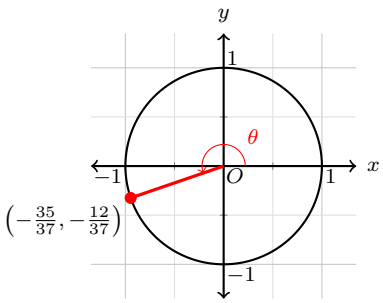
$\tan\theta =$ _____

$\csc\theta =$ _____

$\sec\theta =$ _____

$\cot\theta =$ _____

b.



$\sin\theta =$ _____

$\cos\theta =$ _____

$\tan\theta =$ _____

$\csc\theta =$ _____

$\sec\theta =$ _____

$\cot\theta =$ _____

In part a, why is $\csc\theta$ negative? Use the words *terminal point* and *reciprocal*.

Practice: Reciprocals of a special angle outside Quadrant I

a. Describe and correct the error a student made in evaluating the secant of a 135° angle.

> The coordinates of the terminal point on the unit circle are $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$.
>
> $$\sec 135° = \frac{-\frac{\sqrt{2}}{2}}{1} = -\frac{\sqrt{2}}{2} \quad \times$$

b. Find the secant, cosecant, and cotangent of a 135° angle.

Hint: Quadrant? Reference angle? Terminal point? Then take reciprocals.

$\sec 135° =$ _____ $\csc 135° =$ _____ $\cot 135° =$ _____

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
6H8 ★	Evaluate trigonometric ratios using reference angles: sin, cos, tan	Quiz 1; Quiz 3 bonus (Example 4)
PAE ★	Evaluate trigonometric ratios using reference angles: sec, csc, cot	Quiz 4; Quiz 3 bonus (Example 5)
ZF7	Determine trigonometric ratios using points on the unit circle (optional)	Try It 5 only: six functions from a terminal point; no scored quiz item
NUF	Special right triangles	Warm-up: the 45° and 30°-60° legs
BRP	Reference angles	Warm-up: step 2 of Example 4

Exit question: Practice 19 (Error Analysis)

If the coordinates of the terminal point of an angle θ on the unit circle are $\left(-\frac{3}{5}, \frac{4}{5}\right)$, describe and correct the error a student made in finding $\tan\theta$.

$$\tan\theta = \frac{-\frac{3}{5}}{\frac{4}{5}} = -\frac{3}{4} \quad \times$$

The terminal point is $(x, y) = \left(_____, _____\right)$, in Quadrant _____. The student divided _____ by _____, which gives the _____ of the tangent (the cotangent). The correct value is $\tan\theta = \frac{y}{x} =$ _____.

Before the quiz, finish these sentences.

The terminal point of θ on the unit circle is $\left(_____, _____\right)$.

The reference angle is the acute angle between the terminal side and the _____.

The _____ of the terminal side tells me the signs of $\sin\theta$ and $\cos\theta$.

$\csc\theta$, $\sec\theta$ and $\cot\theta$ are the _____ of $\sin\theta$, $\cos\theta$ and $\tan\theta$.